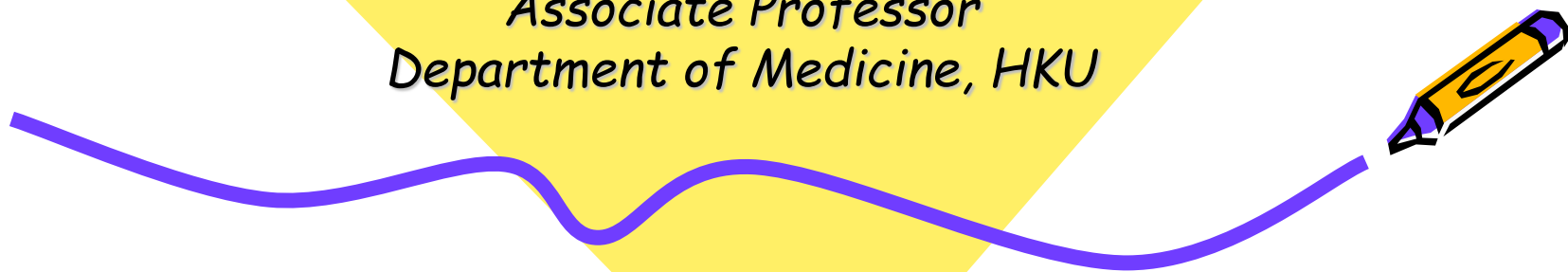


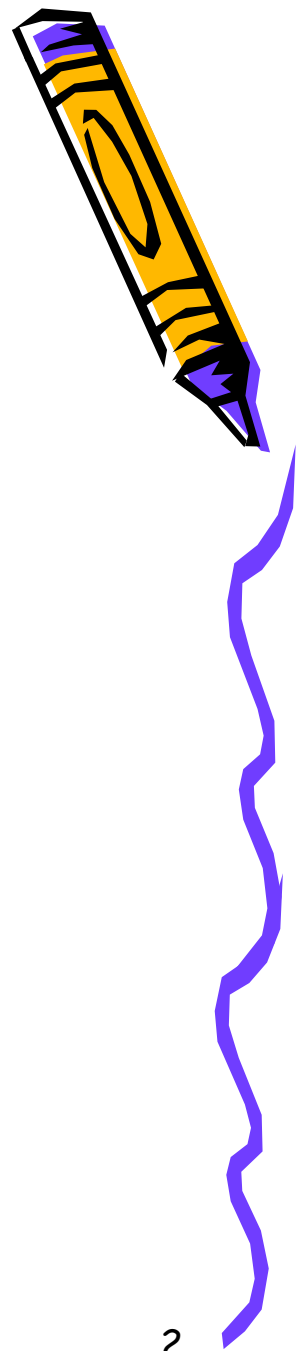


Fever and purulent sputum

*Dr. James C.M. Ho
Associate Professor
Department of Medicine, HKU*



Lower respiratory tract infections



- Acute bronchitis
- Acute exacerbation of chronic bronchitis
- Bronchiectasis
- Pneumonia
- Lung abscess



Case scenario

- M/35, chronic smoker
- Good past health
- Complained of high fever for 3 days
- URTI symptoms (sore throat, running nose) 10 days ago
- Increasing cough and greenish sputum over the past 5 days



Case scenario

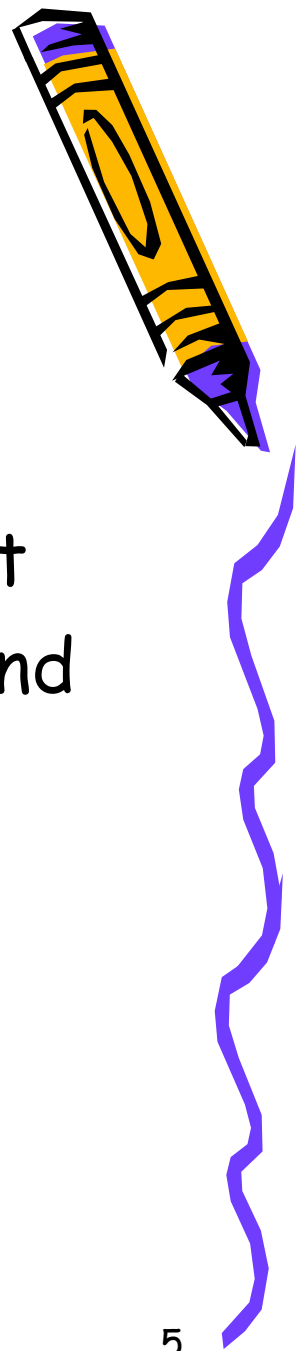


- Progressive shortness of breath with right pleuritic chest pain for 3 days
- High swinging fever with chills and rigor for 3 days
- Seen by GP, given oral antibiotics and paracetamol
- Admitted through A&E for worsening of general condition



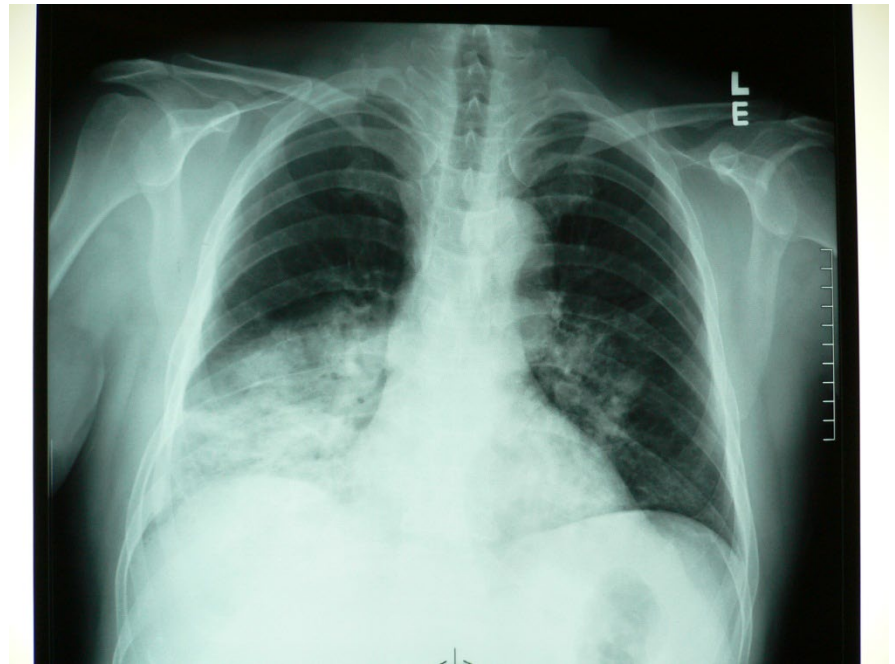
Case scenario

- P/E:
 - in respiratory distress, mentally alert
 - Chest: dullness, bronchial breath sound and coarse inspiratory crackles over right lower chest
 - BP 90/50 mmHg
 - SpO₂ 92% (RA)

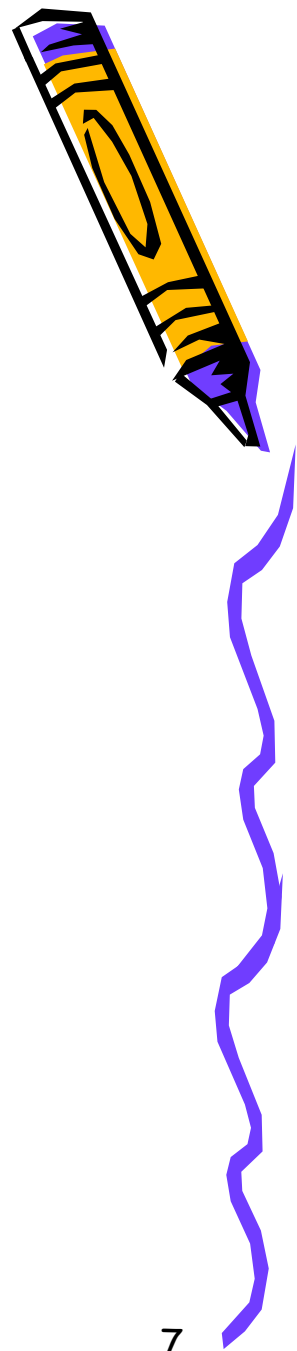


Clinical diagnosis

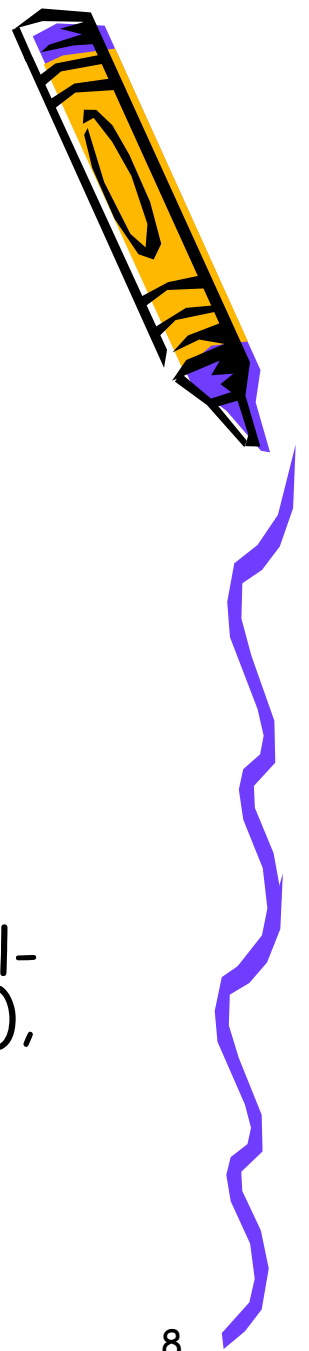
- Community-acquired pneumonia



Pneumonia



Pneumonia: definition



- Inflammation of lung parenchyma, commonly due to infective agents (bacteria, viruses or fungi)
- Classifications:
 - **Anatomical**: lobar pneumonia, multi-lobar, bronchopneumonia
 - **Aetiological**: bacterial, viral, fungal
 - **Clinical**: community-acquired (CAP), hospital-acquired (HAP), ventilator-associated (VAP), aspiration

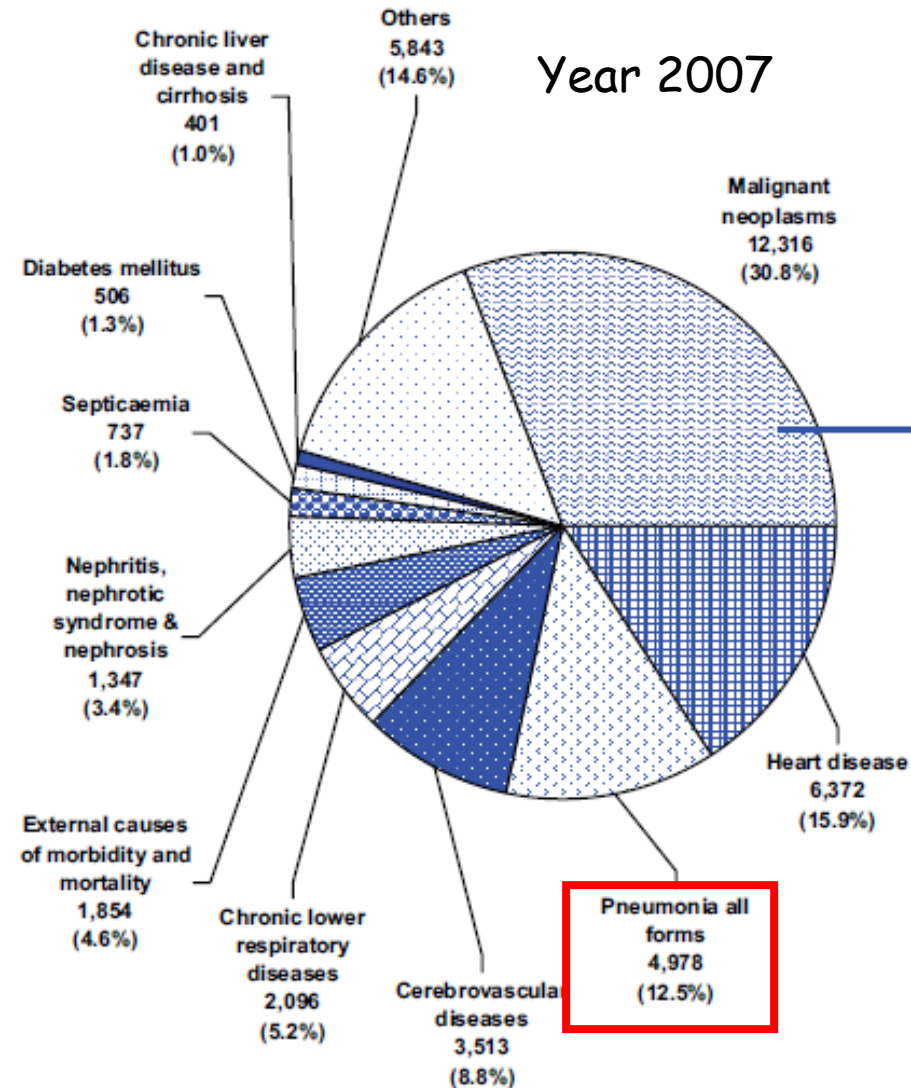




Year 2007

Pneumonia: disease burden

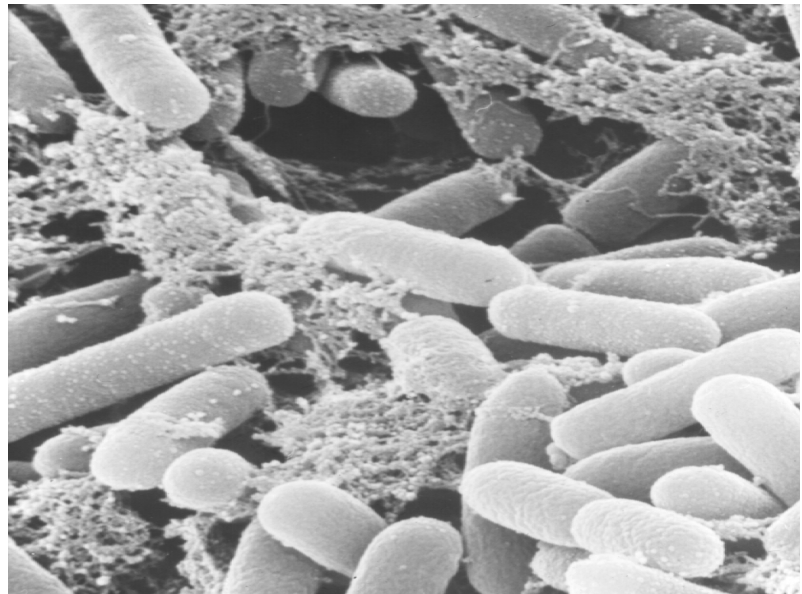
- 12.5% of deaths in 2007 (17% in 2022)
- 3rd leading cause of death since 2003 till 2011, 2nd leading cause since 2012
- 42,403 hospital admissions (2.34% of total admissions) in 2021



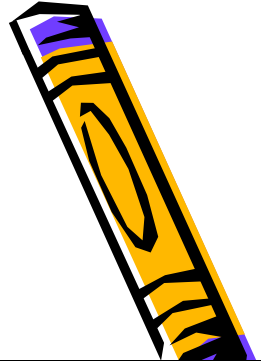
Hospital Authority and DoH, HK

Causative pathogens

- Depends on clinical settings and host factors
- Bacteria
- Viruses
- Fungi
- *Mycobacterium tuberculosis* (esp. in HK or endemic areas)

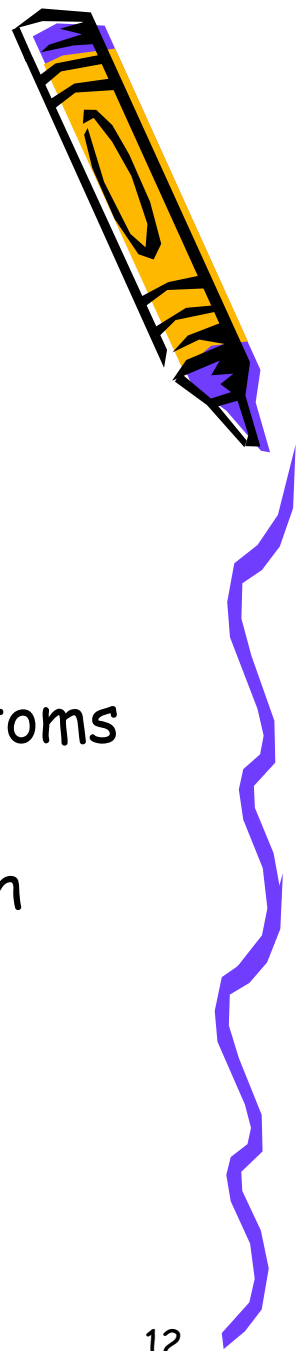


Causative pathogens



		CAP	HAP	Aspiration
Gram +	Streptococcus pneumoniae	++	+	-
Gram +	Haemophilus influenzae	+	+	-
Atypical	Mycoplasma pneumoniae	+	-	-
Atypical	Chlamydophila pneumoniae	+	-	-
Atypical	Legionella pneumophila	+	-	-
Gram -	Klebsiella pneumoniae, Pseudomonas aeruginosa	+/-	++	+
Gram +	Methicillin-resistant Staphylococcus aureus (MRSA)	+/-	++	-
Anaerobes	Bacteroides species	-	+	++

Community-acquired pneumonia (CAP): clinical features

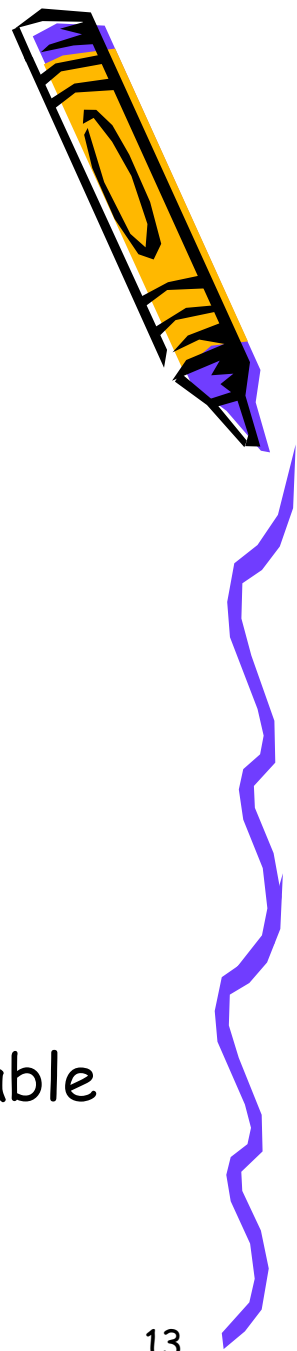


- Symptoms:
 - Cough, sputum, haemoptysis, dyspnoea, pleuritic chest pain
 - Fever, chills, rigor
 - Confusion, constitutional or systemic symptoms
- Signs:
 - Chest: percussion dullness, bronchial breath sound, coarse crackles, increased vocal resonance
 - Vital signs: BP, SpO₂; mental state



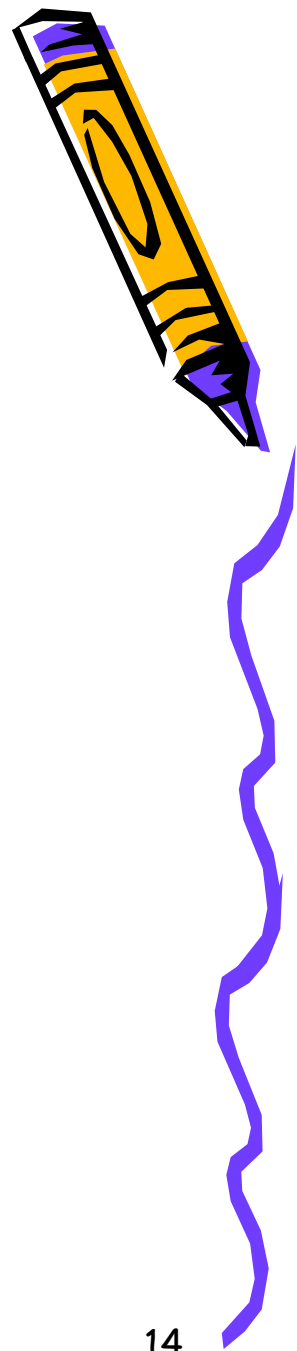
CAP: clinical features

- TOCC
 - Travel
 - Occupation
 - Contact
 - Clustering
 - Atypical pneumonia
 - Usually refers to *M. pneumoniae*, *C. pneumoniae*, *L. pneumophila*
 - Syndrome of pneumonitis, fever, and a relatively normal WBC, without identifiable bacterial pathogens
- Prominent systemic complaints



CAP: clinical approach

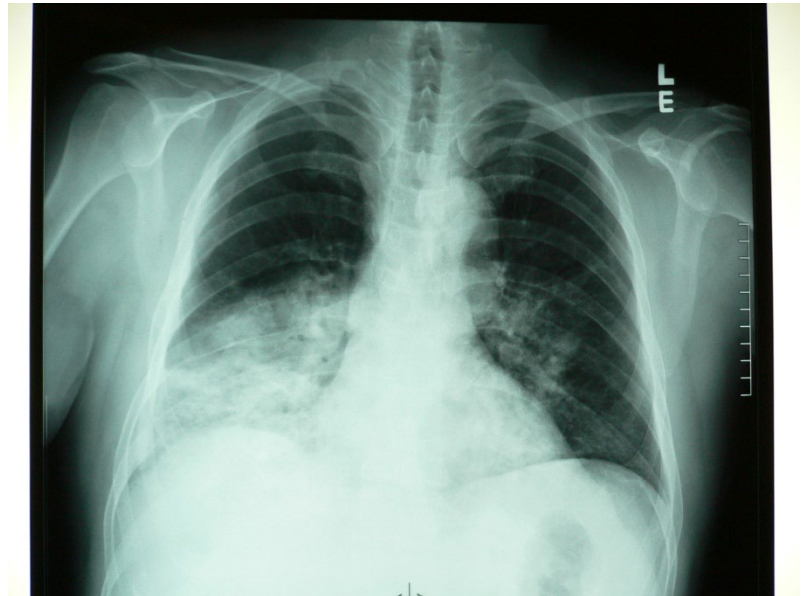
- Establish the clinical diagnosis of pneumonia
- Identify causative pathogens
- Assess clinical severity
- Empirical anti-infective/antibiotic treatment



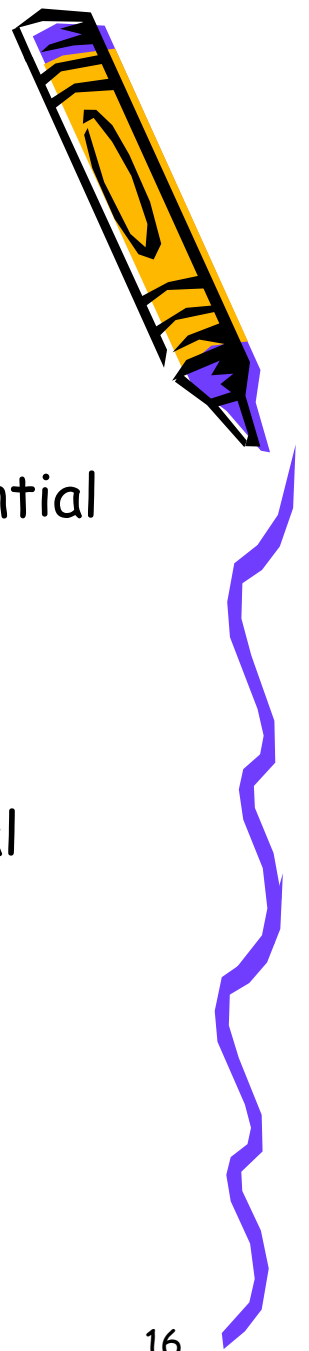
Diagnosis of CAP



- Compatible respiratory symptoms and signs
- Chest xray
 - Consolidations in lobar pneumonia
 - Patchy infiltrates in bronchopneumonia



Diagnosis of CAP



- Further investigations
 - Complete blood counts (total WBC and differential counts)
 - Electrolytes, liver and renal function tests
 - Arterial blood gases
 - Atypical pneumonia serology
 - Nasopharyngeal aspirate if influenza or atypical pneumonia suspected
 - Urinary antigen test for legionella and pneumococcus
 - Sputum and blood cultures



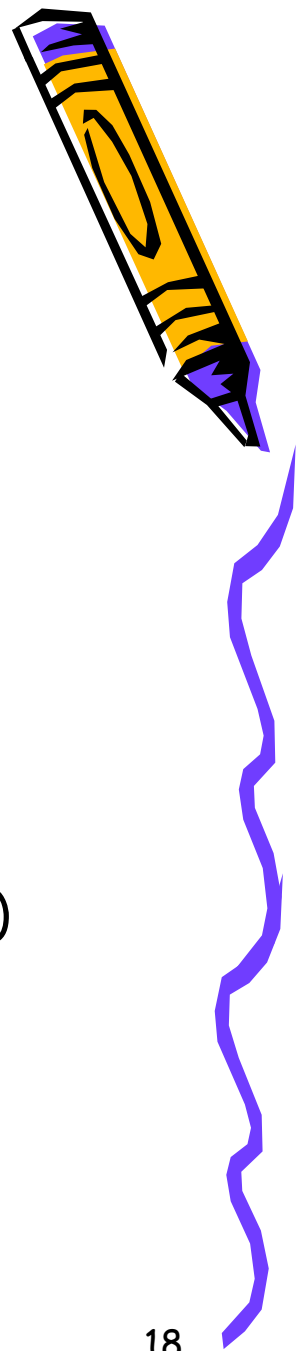
Identify causative pathogens



- Microbiological diagnosis is **retrospective**
- Initial management is based on clinical setting and host factors
 - Co-morbidities, immunocompetent vs immunocompromised hosts, previous antibiotic use
- Need to cover **Streptococcus pneumoniae** in ALL cases of CAP
- Need to cover **Legionella pneumophila** in SEVERE cases of CAP



Assess clinical severity



- Help to decide on appropriate clinical setting for management
 - Outpatient
 - General medical ward
 - Intensive care unit
- Poor prognostic factors
 - Old age
 - Hypoxaemia ($\text{PaO}_2 < 8 \text{ kPa}$ on room air)
 - High ($> 11 \times 10^9/\text{ml}$) or low total WBC ($< 4 \times 10^9/\text{ml}$)
 - High urea ($> 8 \text{ mmol/L}$)
 - Multilobar involvement
 - Bacteraemia



Assess clinical severity



- **CURB-65**
 - Confusion
 - Urea > 7 mmol/L
 - Respiratory rate > 30 /min
 - Blood pressure (SBP < 90 mmHg or DBP < 60 mmHg)
 - Age > 65
- > 2 factors present $\rightarrow$ hospital admission



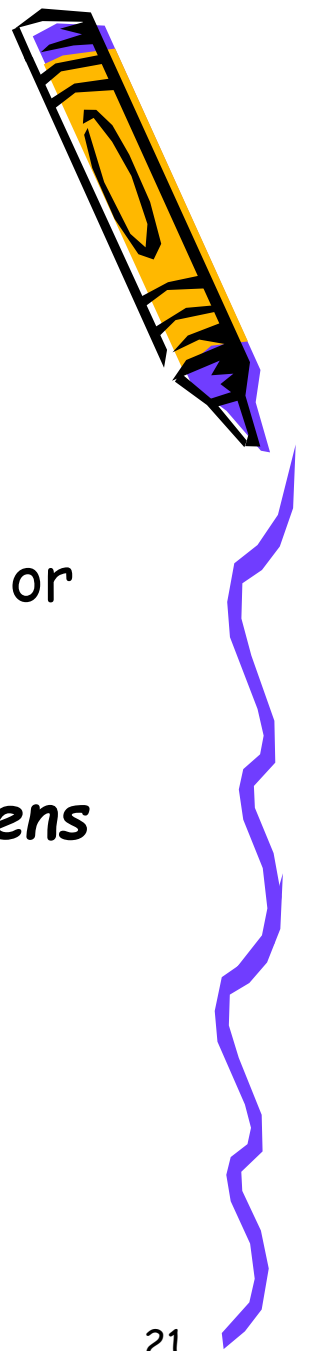
Empirical antibiotics for CAP



- Need to administer appropriate antibiotics timely (preferably **within 6-8 hours**)
- Choice of antibiotics depends on clinical settings and risk factors for multi-resistant pathogens
- Cover **Strep. pneumoniae** and atypical pneumonia (**Legionella** especially in the more severe CAP)



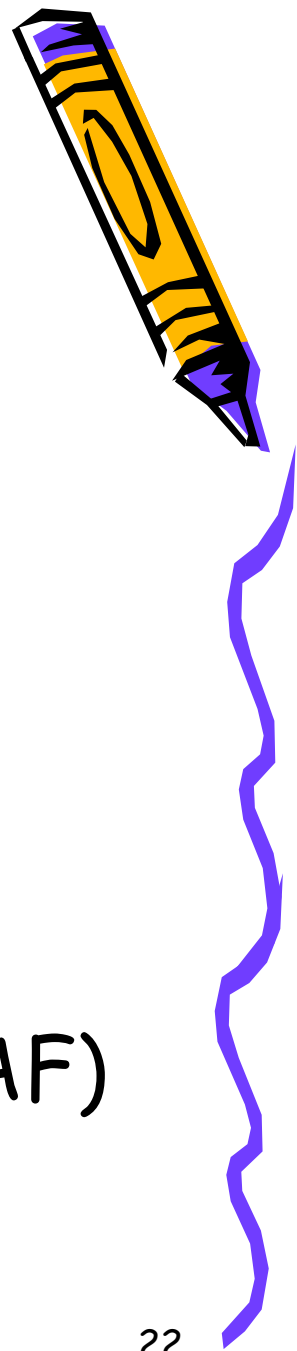
Empirical antibiotics for CAP



- Penicillin with beta-lactamase inhibitor +/- macrolide or tetracycline
- E.g. Amoxicillin-clavulanate +/- clarithromycin or azithromycin or doxycycline
- Duration ~ 7-10 days
- Risk factors for ***multi-drug resistant pathogens***
 - Recent/frequent antibiotic use
 - Recent/frequent hospitalization
 - Nursing home residents
 - Immunosuppressive disease or therapy



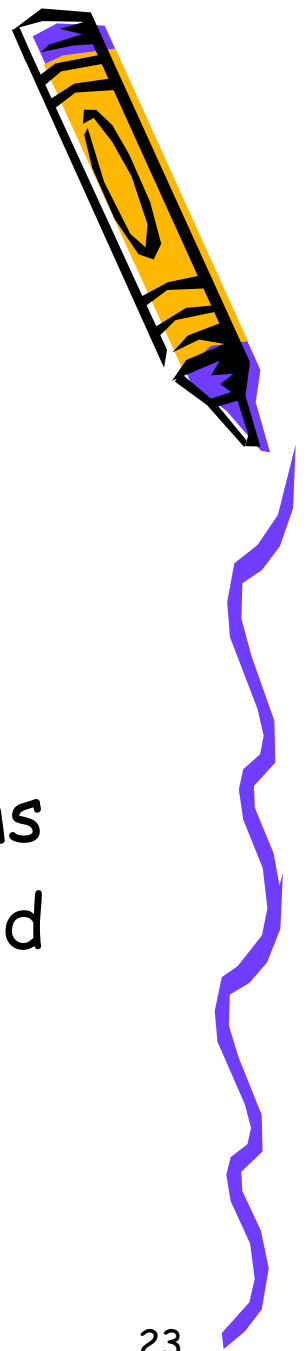
General therapy for pneumonia



- O2 or mechanical ventilation for respiratory failure
- Chest physiotherapy
- Fluid rehydration
- Treat underlying COPD with bronchodilators
- Control cardiac arrhythmias (e.g. AF)



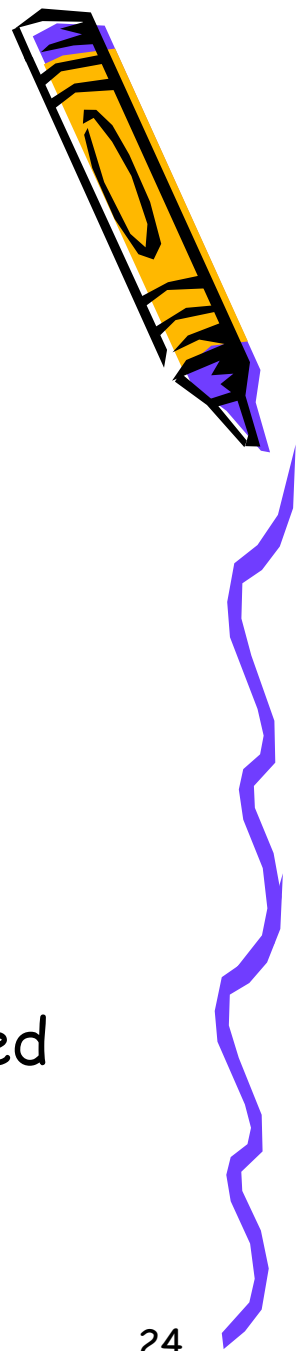
Hospital-acquired pneumonia (HAP): clinical features



- Pneumonia occurring > 48 hrs after hospitalization
- Mostly related to aspiration of oropharyngeal secretions that are colonized by hospital-acquired organisms
- More resistant bacteria (e.g. MRSA) and Gram -ve bacilli anticipated



HAP: clinical features



- Predisposing factors:
 - General debility and old age
 - Cigarette smoking
 - COPD
 - Aspiration
 - Post-G.A.
 - H2 blockers or antacids
 - Nasogastric tube
 - Mechanical ventilation (ventilator-associated pneumonia [VAP])



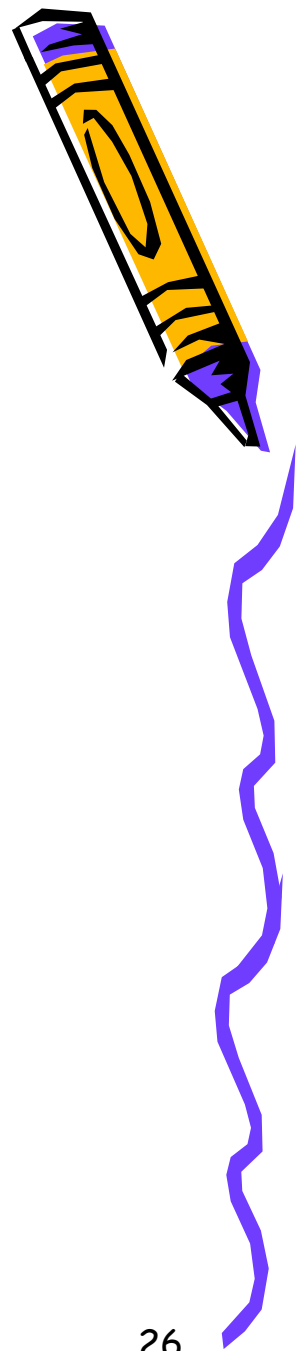
HAP: empirical antibiotics



- Cover Strep. pneumoniae, resistant Gram +ve (may include MRSA if appropriate), Gram -ve (Klebsiella pneumoniae, Pseudomonas aeruginosa, Extended spectrum β -lactamase (ESBL) producing organisms), anaerobes
- Extended spectrum penicillin or third-generation cephalosporin or carbapenem
- E.g. piperacillin-tazobactam or meropenem



Aspiration pneumonia: clinical features



- Predisposing factors
 - Altered consciousness
 - Alcoholism
 - Vomiting
 - Dysphagia and oesophageal diseases
 - Neurological disorders (bulbar palsy)
 - Nasogastric tube feeding
 - Severe dental or upper respiratory tract infections
 - Terminal illness
 - Drowning



Aspiration pneumonia: clinical features



- Usually affecting
**dependent parts of
lungs**
 - Lower lobes (especially
on the right)
 - Posterior segments of
upper lobes (in those
mostly recumbent)
- **Videofluoroscopic
swallowing test
(VFSS)**



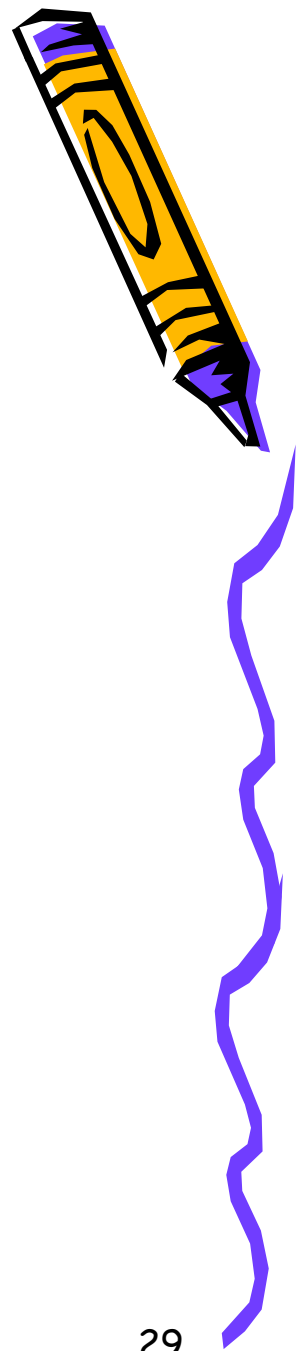
Aspiration pneumonia: management



- Antibiotics
 - Cover oropharyngeal flora especially anaerobes
 - Penicillins
 - E.g. amoxicillin-clavulanate
- Treatment of underlying conditions
- Prevention of further aspiration
 - Non-oral feeding (e.g. via percutaneous endoscopic gastrostomy [PEG])



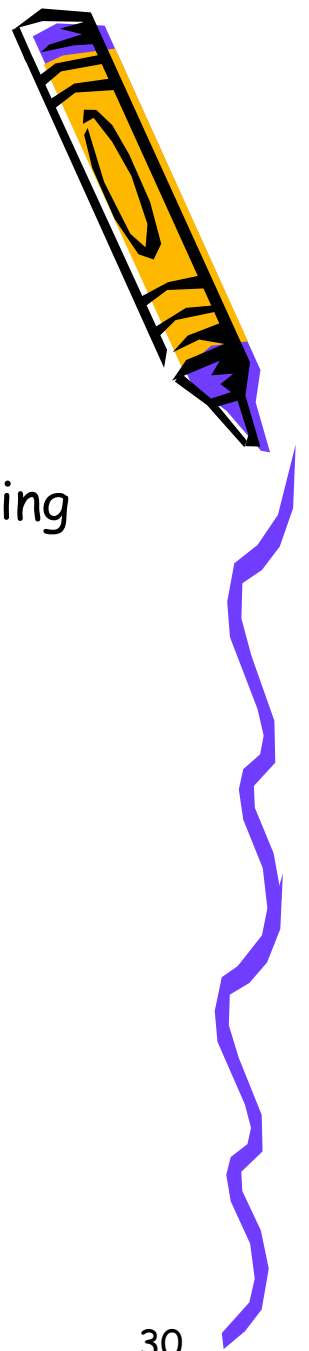
Complications of pneumonia



- Respiratory failure
- Septicaemia with multi-organ failure
- Lung abscess
 - Air-fluid level, cavity
 - Could be polymicrobial
 - Needs prolonged (4-6 weeks) antibiotics
 - No need of percutaneous drainage in most cases
- Parapneumonic effusion



Complications of pneumonia



- Empyema thoracis
 - Pus in pleural cavity (gross appearance, Gram's staining or culture)
 - Needs prolonged (4-6 weeks) antibiotics
 - Pleural drainage (e.g. intercostal tube) is needed
- Electrolyte abnormalities e.g. hypoNa due to SIADH
- Acute myocardial infarction
- Cardiac arrhythmia (AF)



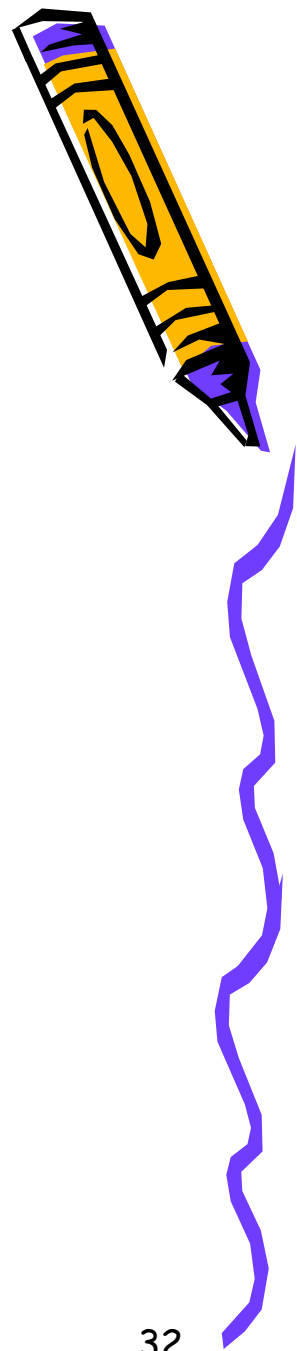
Recurrent pneumonia



- Causes
 - Diffuse bronchopulmonary diseases e.g. bronchiectasis
 - Recurrent aspiration
 - Immunocompromised: HIV, malignancies
 - Bronchial obstruction: tumour, foreign body
- Investigations
 - CT thorax
 - Bronchoscopy
 - Sputum: AFB and cytology
 - Blood: look for immunodeficiency



Influenza



Influenza virus

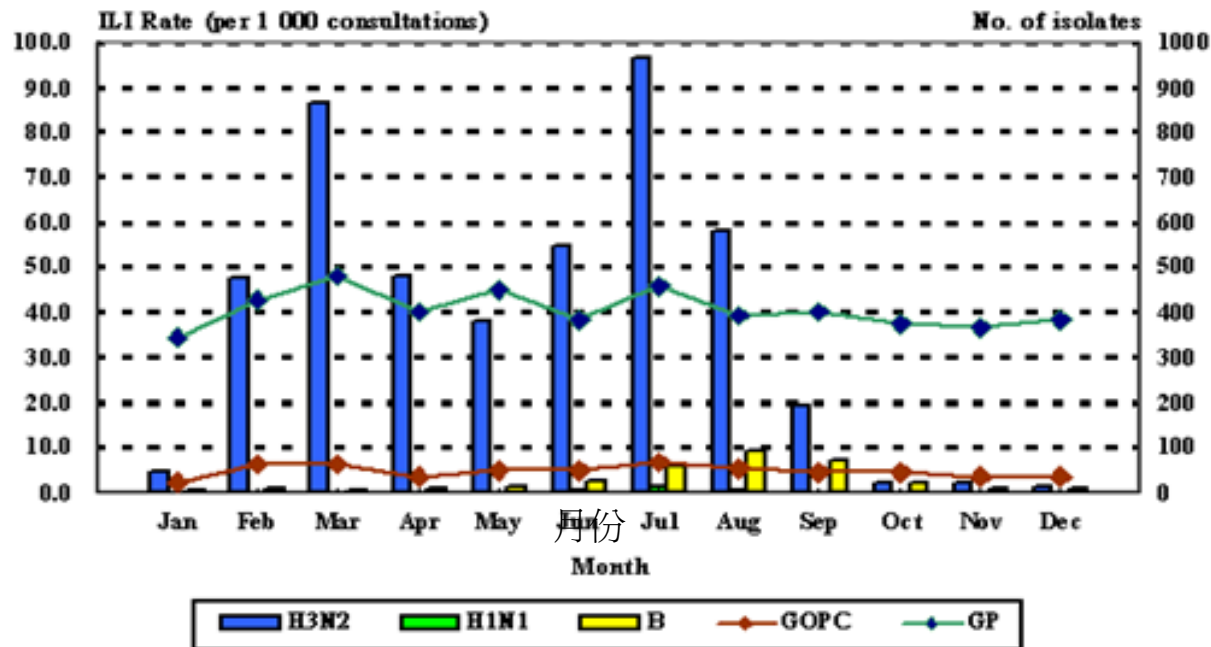


- Influenza A and B
- Immunogenic surface antigens
 - Haemagglutinin (H)
 - Neuraminidase (N)
- *Antigenic shift*: major changes in the antigens (e.g. H3N2 to H1N1)
- *Antigenic drift*: minor changes in the antigens
- Worldwide pandemics due to antigenic shift resulting in a new highly pathogenic virus, without specific natural immunity



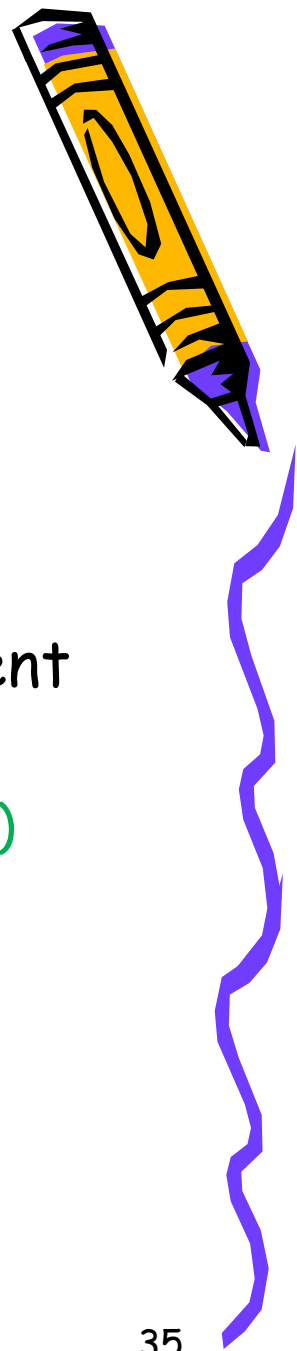
Seasonal Influenza

In Hong Kong, the disease is more common in periods from January to March and from July to August.



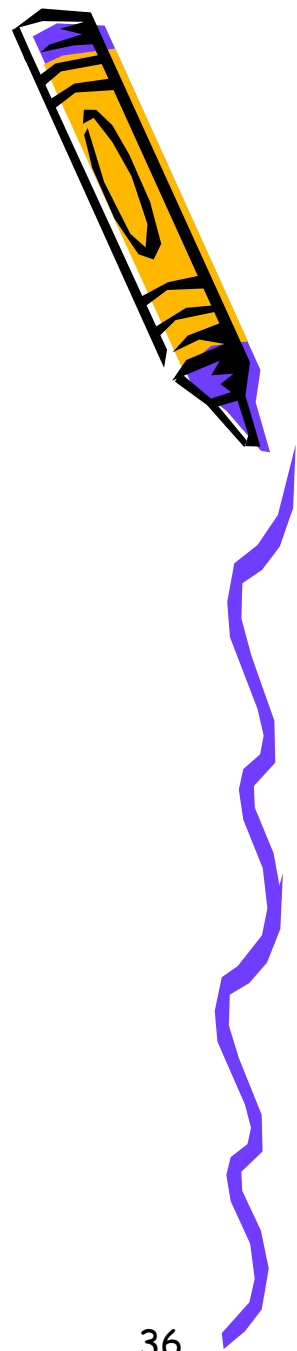
Consultation rate of influenza-like illness and number of influenza isolates in 2004

Clinical features of influenza



- Upper respiratory tract symptoms
 - Running nose, soar throat, cough and sputum
- Fever, myalgia, arthralgia
- Lower respiratory tract or systemic involvement
- Virological diagnosis
 - Rapid antigen test on nasopharyngeal aspirate (NPA)
 - RT-PCR
 - Viral cultures
 - Serology





Management of influenza

- Personal hygiene
- Symptomatic treatment
- Specific antivirals
 - Amantadine and rimantadine
 - Neuraminidase inhibitors



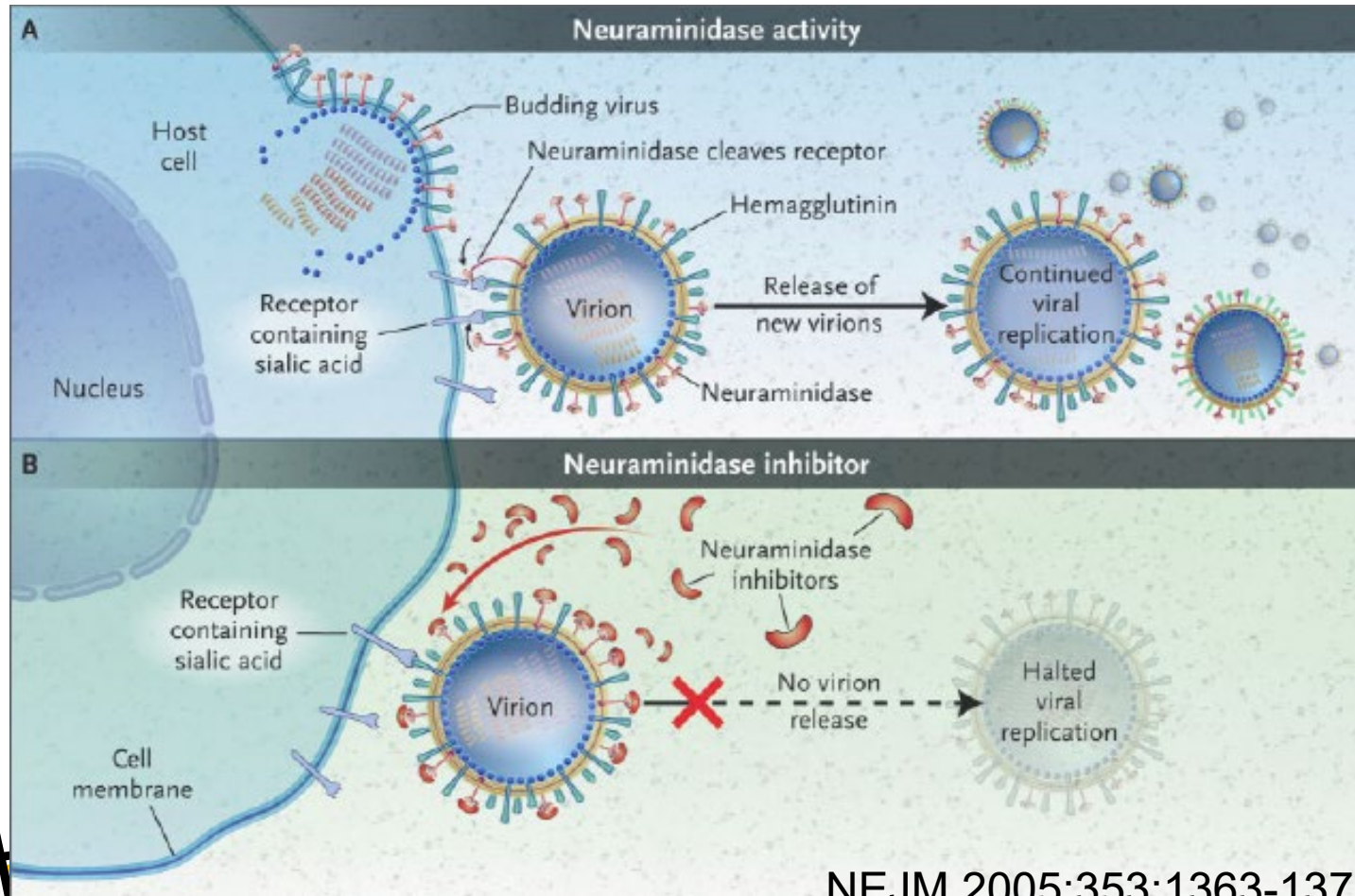
Amantadine and Rimantadine



- Interfere with viral uncoating inside cell
- Effective only against influenza A
- Severe toxic effects, mainly neurological
- Rapid emergence of drug resistance
- Not useful in treating avian flu

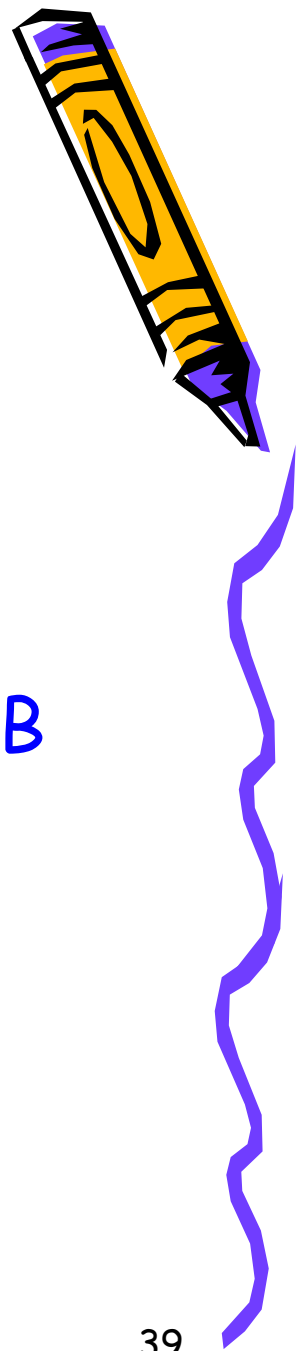


Neuraminidase inhibitors



NEJM 2005;353:1363-1373

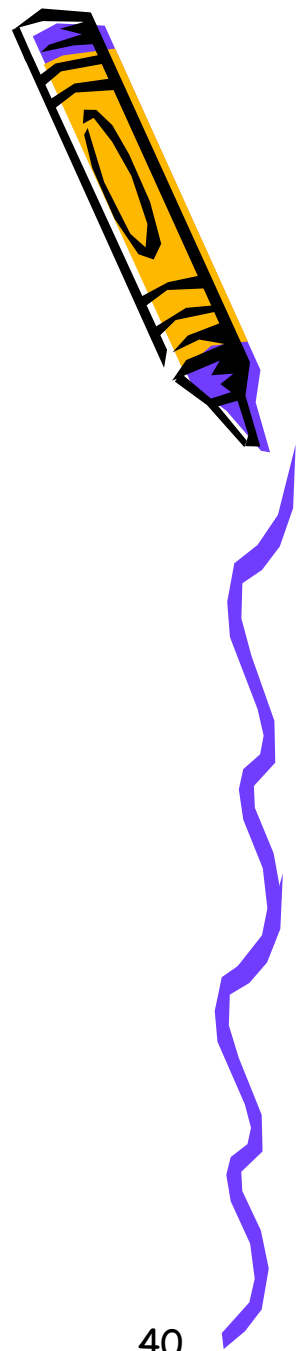
Neuraminidase inhibitors (NI)



- Interfere with release of progeny influenza virus from infected host cells thus stopping spread of infection in respiratory tract
- Effective against both influenza A and B
- Active when given as early as possible (within 48 hours)
- E.g. oseltamivir (Tamiflu), zanamivir (Relenza)



Bronchiectasis



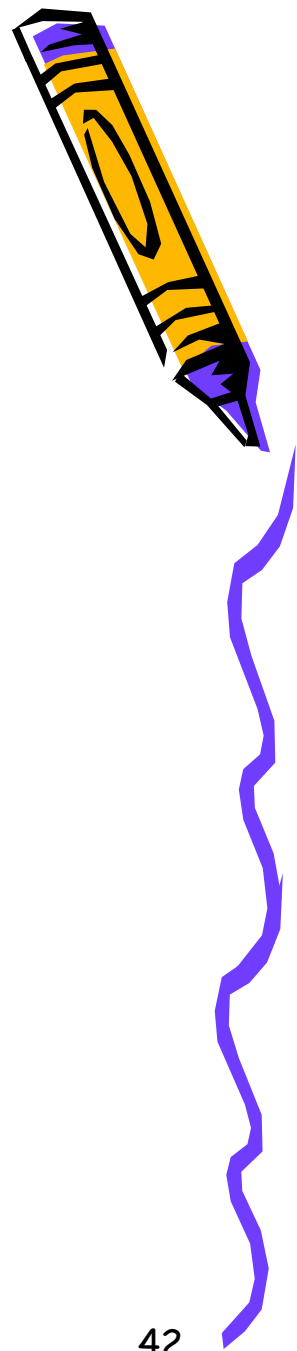
Bronchiectasis: clinical features



- Pathological dilatation of bronchi
- Generalized or localized
- Cause: mostly idiopathic (needs to exclude other causes e.g. post-infection, TB)
- Symptoms
 - Chronic sputum production and cough
 - Infective exacerbations
 - Haemoptysis - usually infective
 - Weight loss, fatigue, chest pain and dyspnoea



Bronchiectasis: clinical features



- Signs

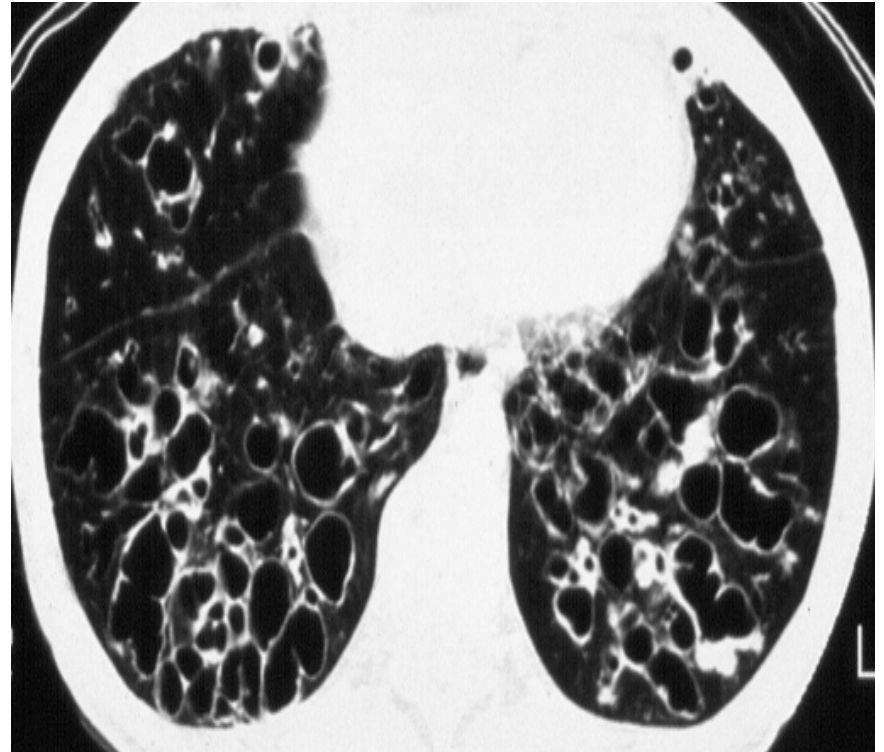
- Cachetic, clubbing, coarse crackles
- Associated COPD
- Respiratory failure
- Complications
- Progressive lung damage
- Lung abscesses
- Empyema
- Brain abscesses



Bronchiectasis: investigations

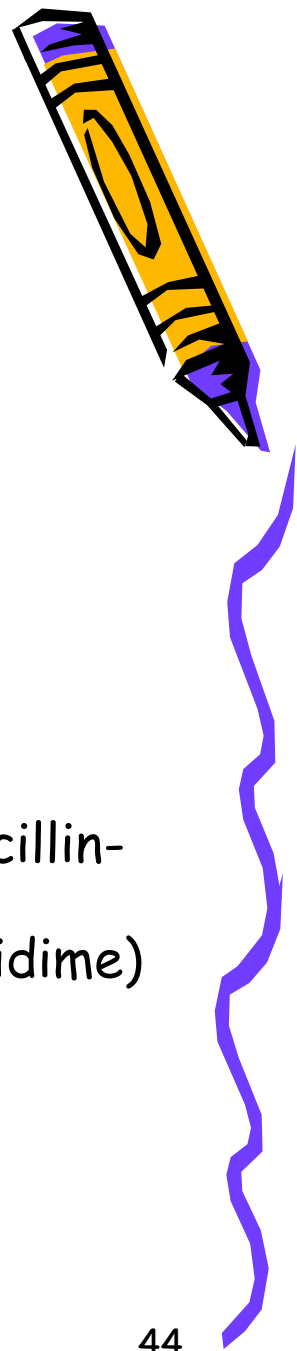


- CXR may be normal or tramline shadows
- HRCT thorax
- Immunoglobulins
- Auto-antibodies
- Barium studies or 24h oesophageal pH monitoring
- Ciliary and sperm analysis
- Sputum culture and AFB
- Neutrophil functions
- Sweat test

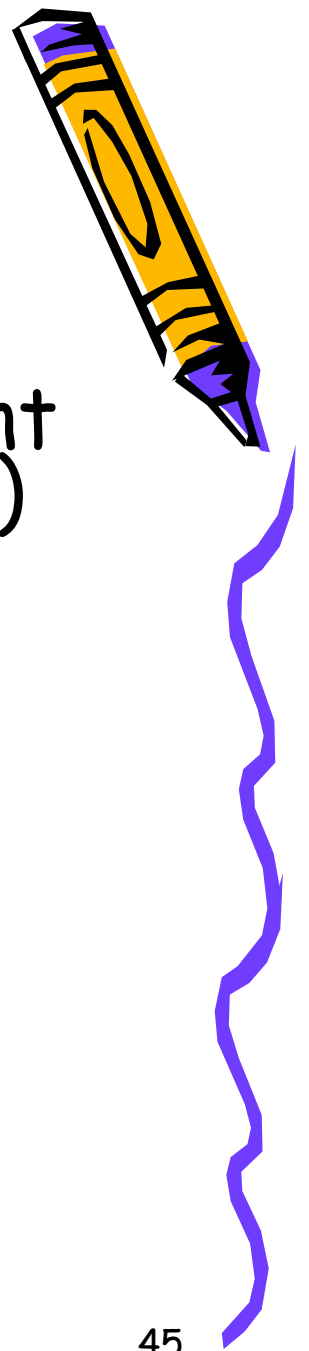


Bronchiectasis: treatment

- Acute exacerbations:
 - Potential organisms
 - Haemophilus influenzae
 - Staphylococcus aureus
 - Pseudomonas aeruginosa (if long-standing)
 - Anti-pseudomonal antibiotics
 - Extended spectrum penicillins (e.g. Piperacillin-tazobactam)
 - 3rd generation cephalosporins (e.g. Ceftazidime)
 - Carbapenems (e.g. Meropenem)
 - Aminoglycosides (e.g. Amikacin)
 - Fluoroquinolones (e.g. Levofloxacin)



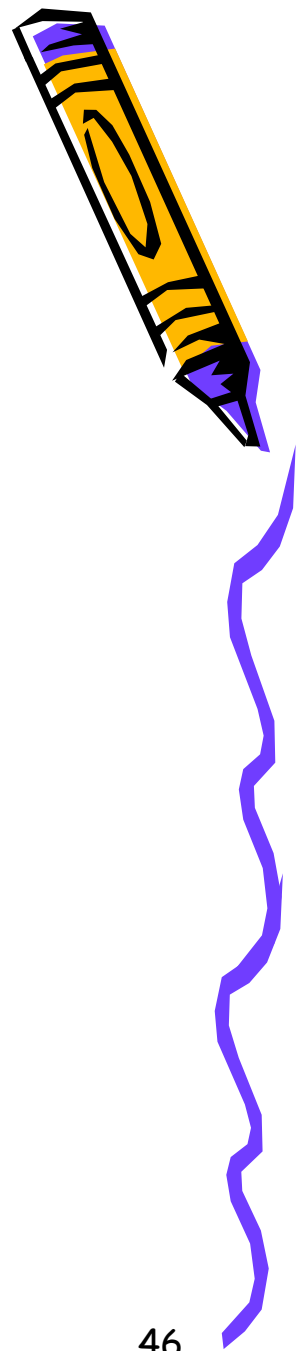
Bronchiectasis: treatment



- ICS +/- LABA (only for concomitant airflow obstruction/bronchospasm)
- Long-term macrolides
 - Immunomodulating rather than antibiotic effect
 - E.g. Azithromycin, erythromycin
 - Watch out for prolonged QTc
- Postural drainage
- Surgery for very limited disease
- Lung transplantation



Learning points



- **Pneumonia**
 - Common pathogens in different settings
 - Clinical features
 - Diagnostic approach and investigations
 - Empirical antibiotic treatment
 - Complications
- **Influenza**
 - Clinical features and management
- **Bronchiectasis**
 - Clinical features and management



TELL US WHAT
JC Whole Class Session
YOU THINK

THANK YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)